

Chapter 1

Bayesian Model

1.1 Statistical model

A *statistical model* M is a family of probability distributions for an observable, $\tilde{y}$, on an observable space, $\tilde{y} \in S_{\tilde{y}}$. For brevity, the sample space will not be made explicit.

A statistical model is *parametric* if it can be indexed by a finite dimensional parameter, θ , that is unknown but belongs to a known parameter space Ω :

$$M = \{p(\tilde{y}|\theta); \theta \in \Omega\},$$

where $\tilde{y}$ represents a future unobserved value. Once the observable has been realized, it is labeled y .

A statistical model is *non-parametric* if the set M contains so many probability distributions, that it can not be indexed by a finite-dimensional parameter. This course focuses on parametric statistical models.

The statistical model M is considered to be “well specified” or “correct” if the true probability distribution $p(\tilde{y}|\theta^*)$ generating the data y belongs to M .

In statistical inference, the observable has already been observed and a given model, M , is assumed to be correct. The goal is to guess which $p(\tilde{y}|\theta^*)$ within M generated the observed data, where $\theta^* \in \Omega$ denotes the “true value” of the parameter.

When a sample of n observations, $\tilde{y} = (\tilde{y}_1, \dots, \tilde{y}_n)$, is obtained and individual obser-

variations $\tilde{y}_i$ are independent and identically distributed (i.i.d.) according to $p(\tilde{y}_i|\theta)$, the statistical model is:

$$M = \left\{ \prod_{i=1}^n p(\tilde{y}_i|\theta); \theta \in \Omega \right\}.$$

This assumes that the same $p(\tilde{y}_i|\theta^*)$ in M is responsible for generating all the $\tilde{y}_j$, which is an assumption that may not always hold in practice.

1.2 Examples of statistical model

Example 1: One throws a pin $n = 30$ times and counts how many times it falls point up, $\tilde{y}$, or tosses a coin $n = 30$ times and counts heads, $\tilde{y}$, or samples $n = 30$ individuals in Barcelona and checks how many have blue eyes, $\tilde{y}$. The goal is to learn about the probability that any of these three events happens, θ . The statistical model for $\tilde{y}$ is:

$$M = \{\text{Binomial}(n = 30, \theta); \theta \in (0, 1)\}.$$

Example 2: Let $\tilde{y} = (\tilde{y}_1, \dots, \tilde{y}_n)$ be the number of goals scored by Girona FC in soccer league games over a season. The purpose is to learn about the expected number of goals scored per game, θ . The statistical model is:

$$M = \left\{ \prod_{i=1}^n \text{Poisson}(\theta); \theta \in (0, \infty) \right\}.$$

Example 3: Let $\tilde{y} = (\tilde{y}_1, \dots, \tilde{y}_n)$ be the lifespan of n black slugs. The goal is to learn about their expected lifespan $1/\theta$, using the statistical model:

$$M = \left\{ \prod_{i=1}^n \text{Exponential}(\theta); \theta \in (0, \infty) \right\}.$$

Example 4: Each face of a fair die is either black or white, but we are not told how many of its faces are black. One rolls the die $n = 30$ times and counts the times the uppermost face is black, $\tilde{y}$. The goal is to learn about the number of black faces of the die. Here the statistical model for $\tilde{y}$ has discrete parameter space:

$$M = \{\text{Binomial}(n = 30, \theta_i); \theta_i \in \{0, 1/6, \dots, 5/6, 1\}\}.$$

Example 5: When comparing two populations, $j = 1, 2$, through two independent samples of observations, $\tilde{y}_1 = (\tilde{y}_{11}, \dots, \tilde{y}_{1n_1})$ and $\tilde{y}_2 = (\tilde{y}_{21}, \dots, \tilde{y}_{2n_2})$, that are i.i.d., the statistical model for each sample usually is:

$$M_j = \left\{ \prod_{i=1}^{n_j} p(\tilde{y}_{ji} | \theta_j); \theta_j \in \Omega \right\}, \text{ for } j = 1, 2,$$

assuming that the two models are the same but the ‘true’ value of the parameter, θ_1 and θ_2 , can differ. In that case the statistical model for both samples is:

$$M = \left\{ \prod_{i=1}^{n_1} p(\tilde{y}_{1i} | \theta_1) \prod_{i=1}^{n_2} p(\tilde{y}_{2i} | \theta_2); (\theta_1, \theta_2) \in \Omega \times \Omega \right\},$$

and the goal will be to learn about $\theta_2 - \theta_1$ or θ_2/θ_1 .

Four special cases of Example 5 that will be considered are the ones:

- **Example 5.1:** comparing the probability of two different pins falling point up, through Binomial models, or
- **Example 5.2:** comparing the expected number of goals scored in a game played at home and in a game played away, through Poisson models, or
- **Example 5.3:** comparing the expected lifespan of black and brown slugs, through Exponential models, or
- **Example 5.4:** comparing the expected birth weight of babies born from mothers that do not smoke and from mothers that smoke, through Normal models.

Example 6: With *BirthWeight0.csv*, we model the weight of a newborn ($\tilde{y} = bwt$) based on *age* and weight (*lwt*) of the mother, on her *race*, on whether she *smoke*, had hypertension (*ht*), or uterine irritability (*ui*), and on number of times undergoing premature labor (*ptl*), and of first trimester visits to the doctor (*ftv*). For that we use a normal linear model,

$$M = \left\{ \prod_{i=1}^n \text{Normal} \left(E[\tilde{y}_i | \beta, x_i] = \beta_0 + \beta_1 x_{1i} + \dots + \beta_{p-1} x_{p-1i}, \sigma^2 \right); (\beta, \sigma^2) \in \mathbb{R}^p \times [0, \infty) \right\}.$$

Example 7: With *kyphosis.csv*, we model the presence of complications after spinal surgery based on *Age* in weeks, *Number* of vertebrae affected, and position of first vertebrae affected, *Start*. Since $\tilde{y}$ is binary, we use a binary logistic model,

$$M = \left\{ \prod_{i=1}^n \text{Binomial} \left(n_i, \pi(\tilde{y}_i | \beta, x_i) = \frac{e^{\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Number}_i + \beta_3 \text{Start}_i}}{1 + e^{\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Number}_i + \beta_3 \text{Start}_i}} \right); \beta \in \mathbb{R}^4 \right\}.$$

1.3 The three problems in statistics

Statistics handles three fundamental types of problems, all involving statistical models.

- Statistical inference: After observing y , and assuming that a given model $M = \{p(\tilde{y}|\theta); \theta \in \Omega\}$ is well specified (meaning that it contains the true distribution $p(\tilde{y}|\theta^*)$ that generated y), the goal is to answer questions about $\theta^* \in \Omega$.

This includes point and interval estimation, hypothesis testing, model selection and predicting future observations.

- Statistical model checking: After observing y , evaluate whether $M = \{p(\tilde{y}|\theta); \theta \in \Omega\}$ is actually well specified. We seek to determine if the true distribution $p(\tilde{y}|\theta^*)$ that generated y is likely to belong to M .
- Design of experiments: Before observing y , decide from which statistical model $M = \{p(\tilde{y}|\theta); \theta \in \Omega\}$ will the data be coming from.

We deal with statistical data analysis, which focuses on statistical inference and model checking. Consequently we will not cover design of experiments.

1.4 Non-Bayesian data analysis

The approach to statistics that only uses a statistical model will be called non-Bayesian. Three implicit assumptions in non-Bayesian data analysis are that:

1. The assumed statistical model, M , is correct and thus the probability model $p(y; \theta^*)$ that generated the data belongs to M .
2. When the sample $Y = (Y_1, \dots, Y_n)$ is i.i.d., every observation Y_i is supposed to be generated by the same distribution $p(y; \theta^*)$.
3. Nothing distinguishes any two parameter values $\theta_1, \theta_2 \in \Omega$. Consequently, no distribution $p(y; \theta)$ in M is favored over another.

The first two assumptions are also implicit in the Bayesian approach, even though the second will be relaxed in Chapter 3. The third assumption is where Bayesians and non-Bayesians depart.

In the non-Bayesian framework, one typically starts selecting one model fit criteria leading to a point estimate $\hat{\theta}$. After that, one develops confidence intervals, hypotheses tests, prediction intervals, model selection criteria and so on, by proceeding as if the fitted model $p(\tilde{y}|\hat{\theta})$ is the “true” $p(\tilde{y}|\theta^*)$ generating the data.

The non-Bayesian approach is highly heuristic, because there is no single road map for choosing the model fit criteria, and for determining how to use the fitted model. Other than under the normal linear model, non-Bayesian methods are usually justified through asymptotic arguments involving normal approximations, and they are often difficult to implement and interpret.

Furthermore, by treating all $p(\tilde{y}|\theta)$ in M as if they were “equally likely” to be the one generating data, $\tilde{y}$, the non-Bayesian approach often defies common sense.

Example 1: Does it make sense to treat the experiments of dropping n pins, of tossing n coins, or of checking n individuals for blue eyes in the same way? Is it equally likely that all $\theta \in (0, 1)$ may be behind the observables in these three experiments? Does observing $y = 9$ with $n = 30$ tell you the same in each of these three scenarios?

Example 2: If θ is the expected number of goals scored in a soccer game by GironaFC, should all $\theta \in (0, \infty)$ be treated as equally likely? This value is certainly going to be smaller than 10!

Example 3: If $\theta = 1/E(\tilde{y}|\theta)$ is the inverse of the expected lifespan of a black slug, should all $\theta \in (0, \infty)$ be treated as equally likely? Since slug life expectancy is surely under 50 years, θ must be greater than $1/50=0.02$!

1.5 Likelihood based statistics

Given M , R.A. Fisher defined the *likelihood function* to be any function of θ on Ω that is proportional to the probability density functions in M evaluated at the observed y ,

$$L_y(\theta) \propto p(\tilde{y} = y|\theta).$$

The rationale is that the larger the value of $L_y(\theta_i)$, the more likely it is that the distribution $p(\tilde{y}|\theta = \theta_i)$ generated the data. So, after we observe y we can use the likelihood function to rank the $\theta \in \Omega$ from most likely to least likely to have produced y .

The likelihood principle asserts that all the information provided by the data regarding

θ is in the likelihood function. According to that principle, if two different experiments yield the same likelihood function, the statistical conclusions must be identical.

Example 1: Consider the pin throw, coin toss and blue eye sampling scenarios using the binomial statistical model:

$$M = \left\{ p(\tilde{y}|\theta) = \binom{n}{\tilde{y}} \theta^{\tilde{y}} (1-\theta)^{n-\tilde{y}}; \theta \in (0, 1) \right\}.$$

If $n = 30$ and we observe $y = 9$, the likelihood function for the three cases is:

$$L_y(\theta) \propto \theta^y (1-\theta)^{n-y} = \theta^9 (1-\theta)^{21}.$$

The maximum likelihood estimate (mle) is $\hat{\theta}_{mle} = y/n = 0.3$.

Since the likelihood function and the mle are the same for the pin, for the coin and the blue eye sample, should our conclusions be the same?

One might be tempted to “standardize” the likelihood function so that the area under the curve is one,

$$L_y^{stand}(\theta) = \frac{\theta^9 (1-\theta)^{21}}{\int_0^1 \theta^9 (1-\theta)^{21} d\theta},$$

and then interpret it as if it was a probability density function, treating likelihood as if it was probability. The problem is that likelihood functions do not behave as density functions when one changes variable (re-parametrizes), and as a consequence likelihood cannot be interpreted as probability.

1.6 Bayesian model

The Bayesian approach starts with a statistical model, $M = \{p(\tilde{y}|\theta); \theta \in \Omega\}$. However, here the parameter θ is treated as a random variable. Before observing the data, the researcher has to be ready to choose a probability distribution $\pi(\theta)$ on Ω .

The prior distribution $\pi(\theta)$, quantifies the knowledge about θ before data is collected.

A *Bayesian model* is a family of probability distributions ordered from most probable (believable) to least probable (believable), based on the *prior distribution*,

$$M_B = \{p(\tilde{y}|\theta); \theta \in \Omega; \pi(\theta)\}.$$

Note that different from the likelihood function, the prior ranks the $\theta \in \Omega$ before observing data. A Bayesian is someone willing to choose a prior. If there is information about θ available, why should we ignore it?

Example 1: Consider the pin throw, coin toss and blue eye sampling scenarios using the Binomial Beta model

$$M_B = \{\text{Binomial}(n, \theta); \theta \in (0, 1); \theta \sim \text{Beta}(a, b)\},$$

where the pdf of the $\text{Beta}(a, b)$ distribution is:

$$\pi(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} \propto \theta^{a-1} (1-\theta)^{b-1},$$

with $\theta \in [0, 1]$. To help choose the values for a and b , we can use the fact that:

$$E[\theta] = \frac{a}{a+b},$$

$$V[\theta] = \frac{ab}{(a+b)^2(a+b+1)} = \frac{E[\theta](1-E[\theta])}{a+b+1}.$$

For a fixed $E[\theta]$, the larger $(a+b)$ the smaller $V[\theta]$ and the more “informative” the prior. In our three scenarios, our prior belief might be such that:

1. for the probability that the pin will fall point up, we might use a $\text{Beta}(2, 2)$ prior, with $E[\theta] = 0.5$ and $V[\theta] = 0.05$,
2. for the probability that the coin will fall heads, we might use a $\text{Beta}(100, 100)$ prior, with $E[\theta] = 0.5$ and $V[\theta] = 0.0012$,
3. for the probability that the individual sampled in Barcelona will have blue eyes, we might use a $\text{Beta}(2, 18)$ prior, with $E[\theta] = 0.1$ and $V[\theta] = 0.0043$.

1.7 Prior predictive distribution

By defining a prior distribution for θ , we establish a joint distribution for $(\tilde{y}, \theta)$. From this, we can derive the marginal distribution for $\tilde{y}$,

$$p(\tilde{y}) = \int_{\Omega} p(\tilde{y}|\theta) \pi(\theta) d\theta = E_{\pi}[p(\tilde{y}|\theta)].$$

This is known as the *prior predictive distribution* for $\tilde{y}$.

This distribution is a weighed average of all the probability models $p(\tilde{y}|\theta)$ in M , with weights provided by the prior. It quantifies our expectations about what we are going to observe before data is collected.

Unlike $p(\tilde{y}|\theta)$, the prior predictive distribution does not depend on any unknown parameter, because it is a fully specified probability model.

Example 1: The prior predictive distribution when we plan a sample size of $\tilde{n}$ is:

$$\tilde{y} \sim \text{BetaBinomial}(\tilde{n}, a, b),$$

with probability mass function given by:

$$p(\tilde{y}) = \binom{\tilde{n}}{\tilde{y}} \frac{\Gamma(a+b)\Gamma(a+\tilde{y})\Gamma(b+\tilde{n}-\tilde{y})}{\Gamma(a)\Gamma(b)\Gamma(a+b+\tilde{n})},$$

for $\tilde{y} = 0, 1, \dots, \tilde{n}$, and with:

$$E[\tilde{y}] = \tilde{n} \frac{a}{a+b} = \tilde{n} E[\theta],$$

$$V[\tilde{y}] = \tilde{n} E[\theta](1 - E[\theta]) \frac{(a+b+\tilde{n})}{(a+b+1)}.$$

Consequently, the prior predictive distribution under the tree scenarios is:

1. when dropping the pin $\tilde{n} = 30$ times is $\text{BetaBinomial}(\tilde{n} = 30, 2, 2)$, with $E(\tilde{y}) = 15$ and $V(\tilde{y}) = 51$,
2. when tossing the coin $\tilde{n} = 30$ times is $\text{BetaBinomial}(\tilde{n} = 30, 100, 100)$, with $E(\tilde{y}) = 15$ and $V(\tilde{y}) = 8.582$,
3. when sampling $\tilde{n} = 30$ individuals is $\text{BetaBinomial}(\tilde{n} = 30, 2, 18)$, with $E(\tilde{y}) = 3$ and $V(\tilde{y}) = 6.429$.

1.8 Posterior distribution

Once a statistical model for $\tilde{y}$ and a prior for θ are chosen, and the data is observed, the Bayesian model is updated by replacing the prior distribution by the *posterior distribution*, $\pi(\theta|y)$, obtained through Bayes theorem:

$$\pi(\theta|y) = \frac{p(y|\theta)\pi(\theta)}{p(y)} = \frac{p(y|\theta)\pi(\theta)}{\int_{\Omega} p(y|\theta)\pi(\theta)d\theta} = kp(y|\theta)\pi(\theta),$$

where $p(y)$ is the marginal distribution of $\tilde{y}$ evaluated on the observed y , which is the constant that ensures that $\pi(\theta|y)$ integrates to one.

When computing analytically the posterior distribution, the only complication is the evaluation of this integral in the denominator. If you recognize the posterior distribution by looking at the functional form of $p(y|\theta)\pi(\theta)$, you know the value of this constant without performing the integration.

Note that:

$$\pi(\theta|y) \propto \pi(\theta)p(y|\theta) \propto \pi(\theta)L_y(\theta),$$

and so the prior distribution transforms the likelihood function into a valid probability density function that allows us to compute the probability of any subset of Ω . The standardized version of the likelihood function is the posterior when the prior is constant.

The posterior distribution, $\pi(\theta|y)$, combines the prior information about θ in $\pi(\theta)$, with the information about θ in the data, $L_y(\theta)$. As the size n grows and/or the amount of information in the prior decreases, the posterior becomes more like the likelihood. As the sample size decreases and/or the amount of information in the prior increases, the posterior remains closer to the prior.

All the information available to answer questions about θ is in $\pi(\theta|y)$. Given that data enters the posterior only through the likelihood function, any inference method based on the posterior respects the likelihood principle.

While we begin by computing the posterior analytically, in modern practice we rarely do so. Instead, we use MCMC methods to simulate samples as large as needed from the posterior and posterior predictive distributions.

Example 1: Consider the pin drop, coin toss and blue eye sampling scenarios using the Binomial Beta model

$$M_B = \{\text{Binomial}(n, \theta); \theta \in (0, 1)\}; \theta \sim \text{Beta}(a, b)\}.$$

If we observe y , then,

$$\pi(\theta|y) \propto \pi(\theta)L_y(\theta) \propto \theta^{a+y-1}(1-\theta)^{b+n-y-1},$$

and hence

$$\theta|y \sim \text{Beta}(a+y, b+n-y),$$

with pdf:

$$\pi(\theta|y) = \frac{\Gamma(a+b+n)}{\Gamma(a+y)\Gamma(b+n-y)} \theta^{a+y-1}(1-\theta)^{b+n-y-1}.$$

Below we present the posterior distribution in the usual three scenarios if $n = 30$ and $y = 9$ and one chooses the priors indicated previously.

1. For the probability that the pin falls point up, the posterior is $\text{Beta}(11, 23)$, with $E(\theta|y) = 0.324$ and $V(\theta|y) = 0.0063$,
2. For the probability that the coin falls head, the posterior is $\text{Beta}(109, 121)$, with $E(\theta|y) = 0.474$ and $V(\theta|y) = 0.0011$.
3. For the probability that an individual in Barcelona is blue eyed, the posterior is $\text{Beta}(11, 39)$, with $E(\theta|y) = 0.220$ and $V(\theta|y) = 0.034$.

Choosing a $\text{Beta}(a, b)$ prior for the Binomial model is equivalent to starting the analysis with $n = a + b$ pseudo observations where a are successes and b are failures.

1.9 Posterior predictive distribution

The prior predictive distribution summarizes our knowledge of what is going to be observed before actually observing it. It is what we should use to make predictions before data is gathered.

After observing the data y , we update our knowledge about θ via the posterior distribution. Similarly, we update our knowledge about $\tilde{y}$ via the *posterior predictive distribution*, which is the updated marginal distribution for $\tilde{y}$,

$$p(\tilde{y}|y) = \int_{\Omega} p(\tilde{y}|\theta)\pi(\theta|y)d\theta = E_{\pi(\theta|y)}[p(\tilde{y}|\theta)].$$

This is a weighted average of all probability models $p(\tilde{y}|\theta)$ in M , where the weights are now provided by the posterior distribution instead of the prior.

The posterior predictive distribution is all that a Bayesian needs to answer questions about future values of $\tilde{y}$, after he has observed y .

Since the posterior predictive is a known probability distribution on the observable space, we can use it to simulate synthetic data to try to mimic reality with it. Building a Bayesian model is very much like building a data simulator.

Example 1: Consider the pin drop, coin toss and blue eye sampling scenarios using the Binomial Beta model

$$M_B = \{\text{Binomial}(n, \theta); \theta \in (0, 1); \theta \sim \text{Beta}(a, b)\}.$$

The posterior distribution after observing y is $\pi(\theta|y) = \text{Beta}(a + y, b + n - y)$. If we plan to observe $\tilde{n}$ future Bernoulli realizations next, the posterior predictive distribution is

$$\tilde{y}|y \sim \text{BetaBinomial}(\tilde{n}, a + y, b + n - y).$$

Below we list the posterior predictive distribution in the usual three scenarios if $n = 30$ and $y = 9$ and we choose the priors indicated previously.

1. When dropping the pin again $\tilde{n} = 30$ times it is $\text{BetaBinomial}(\tilde{n} = 30, 11, 23)$, with $E(\tilde{y}|y) = 9.706$ and $V(\tilde{y}|y) = 12.006$.
2. When tossing the coin again $\tilde{n} = 30$ times it is $\text{BetaBinomial}(\tilde{n} = 30, 109, 121)$, with $E(\tilde{y}|y) = 14.217$ and $V(\tilde{y}|y) = 8.419$.
3. When sampling again $\tilde{n} = 30$ individuals it is $\text{BetaBinomial}(\tilde{n} = 30, 11, 39)$, with $E(\tilde{y}|y) = 6.6$ and $V(\tilde{y}|y) = 8.075$.

1.10 Examples of Bayesian model

A family of prior distributions is said to be conjugate for a statistical model, if the resulting posterior distributions belong to the same family as the prior.

When using a conjugate prior, we update the Bayesian model by updating the value taken by the parameters of the prior, bypassing the need to integrate or simulate.

Beta distributions are the conjugate priors for the binomial model.

The first two examples in this section combine the Poisson and the exponential statistical models with their conjugate priors. The last example in this section is special in that the parameter space of the model is finite.

All the calculations in this section are made analytically. In spite of that, there will be advantages in using the MCMC simulated based approach presented later, instead of using the exact approach presented here.

Example 2: Poisson gamma model

The goal is to learn about the expected number of goals scored by Girona FC per league game. The data, $y = (y_1, \dots, y_n)$, is the number of goals scored so far this season. The statistical model is the i.i.d. Poisson model:

$$M = \left\{ p(\tilde{y}|\theta) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{\tilde{y}_i}}{\tilde{y}_i!}; \theta \in (0, \infty) \right\},$$

for $\tilde{y}_i = 0, 1, \dots$, where $E[\tilde{y}_i|\theta] = V[\tilde{y}_i|\theta] = \theta$.

The conjugate priors for this model are Gamma(a, b) with pdf:

$$\pi(\theta) = \frac{b^a}{\Gamma(a)} \theta^{(a-1)} e^{-b\theta}.$$

To choose (a, b) it helps to know that $E[\theta] = a/b$ and $V[\theta] = a/b^2$. Note that for fixed $E[\theta]$, $V[\theta]$ decreases with increasing b .

In our example, we select a Gamma(4, 3) prior, with $E[\theta] = 1.333$ and $V[\theta] = 0.444$.

After observing n games with a total of $\sum_i y_i$ goals scored, the likelihood function is:

$$L_{\sum_i y_i}(\theta) \propto e^{-n\theta} \theta^{\sum_i y_i}.$$

The mle is $\hat{\theta}_{mle} = \bar{y}$, and the standardized likelihood function is:

$$L_y^{stand}(\theta) = \frac{e^{-n\theta} \theta^{\sum_i y_i}}{\int_0^\infty e^{-n\theta} \theta^{\sum_i y_i} d\theta},$$

which would be the posterior under the flat improper prior introduced in the next section.

The pdf of the posterior distribution under a Gamma(a, b) prior is such that:

$$\pi(\theta|y) \propto \theta^{(a+\sum_i y_i-1)} e^{-(b+n)\theta}.$$

Thus, the posterior distribution is Gamma($a + \sum_i y_i, b + n$).

Choosing a Gamma(a, b) prior is equivalent to including in the analysis an extra set of b observations summing a , which in our example would be like GironaFC having scored $a = 4$ goals in $b = 3$ extra games.

In our example, in the first $n = 28$ games we observe $\sum_i y_i = 42$ goals, and therefore the posterior is Gamma(46, 31), with $E[\theta|y] = 1.48$ and $V[\theta|y] = 0.049$.

Under a $\text{Gamma}(a, b)$ prior, the prior predictive distribution for a single observation is $\text{GammaPoisson}(a, b)$, with probability mass function

$$p(\tilde{y}) = \frac{b^a}{\Gamma(a)} \frac{\Gamma(a + \tilde{y})}{(b + 1)^{a + \tilde{y}}} \frac{1}{\tilde{y}!} \quad \text{for } \tilde{y} = 0, 1, \dots$$

where $E[\tilde{y}] = a/b$ and $V[\tilde{y}] = a(b + 1)/b^2$.

The posterior predictive distribution for a single future observation, after observing a sample of n is $\text{GammaPoisson}(a + \sum_i y_i, b + n)$.

In our example the prior predictive distribution is $\text{GammaPoisson}(4, 3)$, with $E[\tilde{y}] = 1.333$ and $V[\tilde{y}] = 1.777$, and the posterior predictive distribution is $\text{GammaPoisson}(46, 31)$, with $E[\tilde{y}|y] = 1.484$ and $V[\tilde{y}|y] = 1.532$.

Example 3: Exponential gamma model

The goal is to learn about the expected lifespan of a black slug, using the lifespans of n black slugs, $y = (y_1, \dots, y_n)$. The statistical model is the i.i.d. Exponential:

$$M = \left\{ p(\tilde{y}|\theta) = \prod_{i=1}^n \theta e^{-\theta \tilde{y}_i}; \theta \in (0, \infty) \right\},$$

for $\tilde{y}_i \in (0, \infty)$, where the expected life length is $E[\tilde{y}_i|\theta] = 1/\theta$ and its variance is $V[\tilde{y}_i|\theta] = E[\tilde{y}_i|\theta]^2 = 1/\theta^2$.

The conjugate priors for the exponential model are again the $\text{Gamma}(a, b)$ with pdf:

$$\pi(\theta) = \frac{b^a}{\Gamma(a)} \theta^{(a-1)} e^{-b\theta},$$

and $E[\theta] = a/b$ and $V[\theta] = a/b^2$.

Given that $\theta = 1/E[\tilde{y}_i|\theta]$ and that we guess the expected lifespans are around 3 years, for this example we select a $\text{Gamma}(3, 9)$ with $E[\theta] = 0.333$ and $V[\theta] = 0.037$.

If we observe $y_1, \dots, y_n$, the likelihood function is

$$L_{\sum_i y_i}(\theta) \propto \theta^n e^{-\theta \sum_i y_i},$$

and the mle is $\hat{\theta}_{mle} = 1/\bar{y}$.

The pdf of the posterior distribution under a $\text{Gamma}(a, b)$ prior is such that

$$\pi(\theta|y) \propto \theta^{(a+n-1)} e^{-(b+\sum_i y_i)\theta}.$$

Thus, the posterior distribution is $\text{Gamma}(a + n, b + \sum_i y_i)$.

Choosing a $\text{Gamma}(a, b)$ prior is equivalent to having an extra set of $a - 1$ observations summing b , which in our example is like having observed two slugs living a total of nine years.

Using the lifespans of $n = 10$ black slugs with $\sum_i y_i = 32.67$ years, we obtain a $\text{Gamma}(13, 41.67)$ posterior, with $E[\theta|y] = 0.312$ and $V[\theta|y] = 0.0075$.

Under a $\text{Gamma}(a, b)$ prior, the prior predictive distribution for a single observation is $\text{GammaExponential}(a, b)$, with probability density function

$$p(\tilde{y}) = \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+1)}{(b+\tilde{y})^{a+1}} \quad \text{for } \tilde{y} \in (0, \infty)$$

where if $a > 1$ then $E[\tilde{y}] = b/(a-1)$ and if $a > 2$ then $V[\tilde{y}] = (ab^2)/((a-1)^2(a-2))$.

The posterior predictive distribution for a single future observation, after having observed a sample of size n , is $\text{GammaExponential}(a + n, b + \sum_i y_i)$.

In our example, the prior predictive distribution is $\text{GammaExponential}(3, 9)$, with $E[\tilde{y}] = 4.5$ and $V[\tilde{y}] = 60.75$, and the posterior predictive distribution is $\text{GammaExponential}(13, 41.67)$ with $E[\tilde{y}|y] = 3.47$ and $V[\tilde{y}|y] = 14.25$.

Example 4: Bayesian model for binary response die

If each face of a fair die is either black or white, but we are not told how many of its faces are black, and if one rolls the die n times and counts the times the uppermost face is black, $\tilde{y}$, we can use as a statistical model for $\tilde{y}$:

$$M = \left\{ p(\tilde{y}|\theta_i) = \binom{n}{\tilde{y}} \theta_i^{\tilde{y}} (1 - \theta_i)^{n-\tilde{y}}; \theta_i \in \{0, 1/6, \dots, 5/6, 1\} \right\}.$$

If $n = 30$ and one observes $y = 22$, the likelihood function is:

$$L_y(\theta_i) \propto \theta_i^y (1 - \theta_i)^{n-y} = \theta_i^{22} (1 - \theta_i)^8,$$

and if the prior is $\pi(\theta_i) = \pi_i$, with $\sum_{i=0}^6 \pi_i = 1$, the posterior distribution is:

$$\pi(\theta_i|y) = \frac{\pi_i \theta_i^y (1 - \theta_i)^{n-y}}{\sum_{i=0}^6 \pi_i \theta_i^y (1 - \theta_i)^{n-y}}, \text{ for } \theta_i \in \{0, 1/6, \dots, 5/6, 1\}.$$

The prior predictive distribution if we intend to observe $\tilde{n}$ new observations is:

$$p(\tilde{y}|y) = \sum_{i=0}^6 \binom{\tilde{n}}{\tilde{y}} \theta_i^{\tilde{y}} (1 - \theta_i)^{\tilde{n}-\tilde{y}} \pi_i,$$

and the posterior predictive distribution is:

$$p(\tilde{y}|y) = \sum_{i=0}^6 \binom{\tilde{n}}{\tilde{y}} \theta_i^{\tilde{y}} (1 - \theta_i)^{\tilde{n}-\tilde{y}} \pi(\theta_i|y), \text{ all for } \tilde{y} = 0, 1, \dots, \tilde{n}.$$

In absence of any prior information on how many faces are black, we will typically choose $\pi(\theta_i) = 1/7$ for all $\theta_i \in \Omega$.

Instead, if we were told that each face was painted black if a fair coin came up heads and it was painted white otherwise, then Bayesians and non-Bayesians alike would agree on using as a prior distribution:

$$\pi(\theta_i) = \binom{6}{i} \left(\frac{1}{2}\right)^6, \text{ for } \theta_i \in \{0, 1/6, \dots, 5/6, 1\}.$$

1.11 Choice of prior distribution

A Bayesian analysis requires the selection of both a statistical model and a prior distribution. Different individuals might choose different priors, but they might also choose different statistical models.

While the choice of a prior is often labeled as "subjective," it is important to recognize that selecting a statistical model is no less so. In fact, getting the statistical model wrong can have worse implications than having a prior centered away from the true θ^* , because the effect of choosing different priors usually washes away when sample size grows, but the effect of choosing different statistical models sometimes doesn't.

The strategy for choosing a prior depends on whether substantive information about the parameters of the model is available before data is incorporated into the analysis.

Subjective prior distributions

If substantive prior knowledge about θ exists, excluding it from the analysis is a missed opportunity for improving the inference. However, translating complex knowledge into a formal prior distribution requires a structured approach.

Choosing a prior distribution for θ implicitly determines a prior predictive distribution for $\tilde{y}$ and vice versa. Therefore, we can choose a prior by either:

1. eliciting a distribution for θ directly, using what we know about θ , or by
2. eliciting a distribution for $\tilde{y}$, using what we know about the observable, then identifying the $\pi(\theta)$ that produces it.

Because translating subjective belief about θ and/or $\tilde{y}$ into a single probability distribution is challenging, we often restrict our search to parametric families. We typically select the specific member of that family by matching first moments or percentiles, and by plotting the pdf of the prior to ensure its shape reflects our beliefs.

When different experts provide conflicting priors, we can either

1. use all priors to reach a consensus prior by combining them into a single mixture,
2. or replicate the analysis with each prior separately and compare conclusions.

It is always useful to carry out a sensitivity analysis trying different priors and checking how the posteriors change. This helps determine how much the final conclusions depend on the initial prior choice.

In the early days of Bayesian statistics, one used conjugate priors for mathematical tractability. With them, posteriors are in the same family as priors, and updating the Bayesian model only requires updating parameter values. Conjugate priors though basically only exist for statistical models that are exponential families, and when they exist they might be too restrictive to reflect complex prior beliefs.

Nowadays we do not rely that much on conjugate priors anymore, because one can resort to MCMC methods to simulate from posterior distributions for any prior deemed appropriate, regardless of its mathematic complexity.

Reference prior distributions

When a researcher is unwilling or unable to elicit substantive prior knowledge about θ , we can resort to “default” “reference” priors. They are designed to introduce as little information as possible about θ , allowing the likelihood to dominate the resulting posterior distribution.

In most cases with sufficient data, the choice among different reference priors becomes practically irrelevant. However, three common approaches exist:

- a) *Flat priors*, which assign equal probability to all $\theta \in \Omega$.

If the parameter space is finite, or a finite interval or a bounded subset of R^p , the posterior distribution is a standardized version of the likelihood function.

When the parameter space is countably infinite or unbounded like in $(-\infty, \infty)$, a flat prior is not a true probability distribution, because it does not integrate to one. The consensus is that one is entitled to use flat priors, as long as one checks that the resulting posterior is a proper distribution that integrates to one.

The limitation of flat priors is that they sometimes lead to improper posterior distributions, and that they are not invariant under re-parametrization and can become highly informative when you do re-parametrize.

A Beta(1, 1) is a U(0, 1) distribution, which is the flat prior for the Binomial model with $\Omega = (0, 1)$. Such a prior is not really non-informative about θ though.

- b) *Jeffrey’s priors*, defined as $\pi(\theta) \propto \sqrt{|I(\theta)|}$, where $|I(\theta)|$ is the determinant of the Fisher information.

The main advantage is that this prior is invariant against re-parametrization, but it is often an improper probability distributions.

The Jeffrey’s prior for the Binomial model is the Beta(1/2, 1/2).

- c) *Diffuse conjugate prior with the largest variance, possibly infinite*.

For the Binomial model this corresponds to a Beta(0, 0), but if we observe $y = 0$ or $y = n$ then the posterior remains improper and inference can not proceed.

Empirical Bayes methods

Empirical Bayesian methods use the observed data itself to select the prior distribution. Specifically, one often selects a prior that results in a prior predictive distribution that

closely matches the empirical distribution of the observed data, y .

This approach often produces statistical methods with excellent non-Bayesian properties, but they are not Bayesian data analysis methods. In Bayesian analysis, the prior distribution must be specified without using the data currently being analyzed.

This does not contradict the fact that in Bayesian practice one uses the posterior distribution, updated in the light of past data, as the prior distribution for the analysis of future observations, in a sequential manner.

1.12 MCMC simulation

MCMC implementation

For Bayesian models that do not use conjugate priors, the analytical approach to finding the posterior is often mathematically intractable. Instead of doing that, we rely on Markov Chain Monte Carlo (MCMC) methods. MCMC allows us to generate a sample of dependent observations from the posterior distribution $\pi(\theta|y)$, virtually for any statistical model and for any prior.

With a simulated sample of θ_i from the posterior $\pi(\theta|y)$, obtaining a sample from the posterior predictive distribution $p(\tilde{y}|y)$, is straightforward. We simply simulate a value y_i from the model $p(\tilde{y}|\theta_i)$ for each θ_i in our sample.

Because we can generate samples as large as we wish from $\pi(\theta|y)$ and $p(\tilde{y}|y)$, we can estimate these distributions and any functional of them, with any degree of precision.

MCMC methods are useful even in conjugate settings where analytical solutions are feasible, because having a large sample from the posterior allows us to do inference about any functional of interest.

MCMC implementations typically involve:

1. discarding the first simulations (burn in stage) to make sure that the Markov chain has attained stationarity,
2. “thinning” the chains by keeping only certain percentage of the simulations left,

- to weaken the dependency of the simulated sample (and save storage space),
3. starting and monitoring several independent chains, and if they all converge to the same stationary distribution, mixing the simulations that we retain from each chain in the final simulated sample.

The convergence of the chains is monitored with trace plots.

Software

In this course we will implement MCMC through JAGS, (“Just another Gibbs sampler”), which can be downloaded from: <https://sourceforge.net/projects/mcmc-jags/files/JAGS/4.x/>

We interface with JAGS from R with the help of the function `jags()` from the `R2jags` library, that needs to be downloaded and installed.

The primary arguments of this `jags` function are:

1. `data`: The observed data, formatted as an R list.
2. `parameters.to.save`: A list of the parameters you wish to monitor and estimate.
3. `model`: A text containing the model in BUGS code (check `jags-distributions.pdf`),
4. MCMC settings `n.iter`, `n.burnin`, `n.thin` and `n.chains`: Indicating number of iterations, of observations to discard at the beginning, amount of thinning, and number of independent chains.

The first three arguments are mandatory. The output of `jags()` is an R list containing the simulated MCMC samples and summary statistics.

1.13 More examples of Bayesian model

All the Bayesian models presented here will be updated using MCMC methods to simulate large samples from the posterior and posterior predictive distributions. In these examples the analytical approach is too complicated to be worth implementing.

When we have prior information about the parameters, we will use it. Otherwise, we will resort to reference priors.

Example 5: Bayesian comparison of two populations

When comparing two populations ($j = 1, 2$), through independent samples, $\tilde{y}_1 = (\tilde{y}_{11}, \dots, \tilde{y}_{1n_1})$ and $\tilde{y}_2 = (\tilde{y}_{21}, \dots, \tilde{y}_{2n_2})$, we typically assume both follow the same structural model:

$$M_j = \left\{ \prod_{i=1}^{n_j} p_j(\tilde{y}_{ji}|\theta_j); \theta_j \in \Omega_j \right\}.$$

While usually $p_1(\cdot|\theta) = p_2(\cdot|\theta)$ and $\Omega_1 = \Omega_2$, the ‘true’ value of the parameter θ_1 and θ_2 , and our prior distributions $\pi_1(\theta_1)$ and $\pi_2(\theta_2)$, may differ. Therefore, the statistical model for both samples is:

$$M = \left\{ \prod_{j=1}^2 \prod_{i=1}^{n_j} p(\tilde{y}_{ji}|\theta_j); (\theta_1, \theta_2) \in \Omega \times \Omega \right\}.$$

The goal is to estimate the joint posterior distribution of (θ_1, θ_2) , and from this distribution obtain the posterior of $\theta_2 - \theta_1$, θ_2/θ_1 or an estimate of $\Pr(\theta_1 > \theta_2|y_1, y_2)$.

Example 5.1: To compare the probability that two different pins fall pointing up, we can use the Binomial Beta model in Example 1 for each pin.

Example 5.2: To compare the expected number of goals scored by GironaFC per games played at home, θ_1 , versus away, θ_2 , we can use a Poisson model. for each situation.

We incorporate substantive knowledge about (θ_1, θ_2) by assuming a $\text{Gamma}(5, 3)$ prior for θ_1 , with $E[\theta_1] = 1,67$, and a $\text{Gamma}(3, 3)$ prior for θ_2 , with $E[\theta_2] = 1$. We also consider a more conservative setting where the same $\text{Gamma}(4, 3)$ prior is chosen for both θ_1 and θ_2 .

Using data from the first 28 games of the 2022/23 season, $n_1 = 14$ games were played at home scoring a total of $\sum_i y_{1i} = 25$ goals, and so $\pi(\theta_1|y_1) = \text{Gamma}(30, 17)$. In the $n_2 = 14$ games played away they scored $\sum_i y_{2i} = 17$ goals, and so $\pi(\theta_2|y_2) = \text{Gamma}(20, 17)$.

What is of interest is the posterior distribution of $\theta_1 - \theta_2$, to quantify what is the home field advantage. That is best explored through MCMC simulation.

Example 5.3: To compare the expected lifespan of black and brown slugs, θ_1 and θ_2 , we use an Exponential model for each type of slug.

We use substantive knowledge about slug lifespans to choose two independent $\text{Gamma}(3, 9)$ priors for θ_1 and θ_2 , with $E[\theta_j] = 1/3$.

With the sample of lifespans used in Example 2, we find that $\pi(\theta_1|y_1) = \text{Gamma}(13, 41.67)$. With a sample of lifespans of $n_2 = 12$ brown slugs with $\sum_i y_{2i} = 13.04$, we find that $\pi(\theta_2|y_2) = \text{Gamma}(15, 22.04)$.

By simulation we can easily estimate the posterior distribution of the difference in mean lifespans, $1/\theta_1 - 1/\theta_2$, to determine which species lives longer on average.

Example 5.4: To compare the expected birth weight of babies born from non-smoking and smoking mothers, we use independent normal models,

$$M_j = \left\{ \prod_{i=1}^n \text{Normal}(\mu_j, \sigma_j^2); (\mu_j, \sigma_j^2) \in \mathbb{R} \times (0, \infty) \right\}.$$

Here we do not want to use substantive prior knowledge about μ_j , and so we assume the μ_j to be independent and:

$$\mu_j \sim \text{Normal}(m_j = 3000, p_j = V[\mu_j]^{-1} = 0.000001),$$

thus centered at $m_j = 3000$ grams, and with a large variance, $V[\mu_j] = 1/p_j = 1000000$.

The $\tau_j = 1/\sigma_j^2$ assumed to be independent $\text{Gamma}(a_j = 0.01, b_j = 0.01)$, with $E[\tau_j] = 1$ and a large variance, $V[\tau_j] = 100$.

Example 6: Bayesian normal linear model

Using data from Hosmer and Lemeshow (1989), in *BirthWeight0.csv*, we aim to model a newborn birth weight (*bwt*), based on the *age* and weight (*lwt*) of the mother, on her *race*, *smoking* status, history of hypertension (*ht*), or uterine irritability (*ui*), and on the number of previous premature labor (*ptl*) and first trimester physician visits (*ftv*).

This is modeled using a normal linear model,

$$M = \left\{ \prod_{i=1}^n \text{Normal}(E[\tilde{y}_i|\beta, x_i] = \beta_0 + \beta_1 x_{1i} + \dots + \beta_{p-1} x_{p-1i}, \sigma^2); (\beta, \sigma^2) \in \mathbb{R}^p \times (0, \infty) \right\}.$$

To minimize the impact of prior beliefs we resort to a reference prior. That is done by assuming a “vague” distribution with large variances, ensuring the likelihood dominates the posterior distribution. We assume that the β_i are a priori independent,

$$\beta_i \sim \text{Normal}(E[\beta_i] = 0, V[\beta_i] = 1000000).$$

The large variance indicates a lack of commitment on any specific range of values for the effect sizes.

As a prior distribution for σ^2 it is assumed that $\tau = 1/\sigma^2$ is independent of the β_i and

$$\tau = 1/\sigma^2 \sim \text{Gamma}(a = 0.01, b = 0.01).$$

This prior has mean 1 and a variance of 100, making it minimally informative.

Example 7: Bayesian binary logistic model

Using data from Chambers and Hastie (1992), in *kyphosis0.csv*, we aim to model the presence of complications after spinal correction surgery. The analysis is based on the *Age* of the patient in weeks, the *Number* of vertebrae involved, and the position of the first topmost vertebra operated on, *Start*.

Since the outcome is binary, we use an additive binary logistic model,

$$M = \left\{ \prod_{i=1}^n \text{Binomial} \left(1, \pi(\tilde{y}_i | \beta, x_i) = \frac{e^{\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Number}_i + \beta_3 \text{Start}_i}}{1 + e^{\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Number}_i + \beta_3 \text{Start}_i}} \right); \beta \in \mathbb{R}^4 \right\}.$$

As in the previous linear regression example, we adopt a non-informative reference prior for the coefficients β . This ensures that our prior beliefs do not unduly influence the posterior results. We assume the β_i coefficients are a priori independent and follow a diffuse Normal distribution:

$$\beta_i \sim \text{Normal}(E[\beta_i] = 0, V[\beta_i] = 1000000).$$

1.14 Bayesian data analysis

In non-Bayesian data analysis the typical workflow involves selecting a statistical model and a model fit criteria, fitting the model to the data, and checking the model through residual analysis. Once a final model is selected, the researcher proceeds “as if” the fitted model was the true probability model that generated the data.

Instead, Bayesian data analysis uses Bayes theorem to update the prior model (statistical model + prior + prior predictive), into the posterior model (same statistical model + posterior + posterior predictive), and then checks it, all through simulation.

Building a Bayesian model is fundamentally like building a data simulator. A Bayesian model is useful only if its posterior predictive distribution can replicate data similar to the data observed and expected to be observed.

Bayesian analysis provides a universal four-step road map for any problem consisting in:

1. choosing a statistical model and a prior distribution,
2. updating that model by computing (or simulating from) the posterior and posterior predictive distribution,
3. checking whether the model is acceptable by checking whether it simulates data resembling real data, improving the model and re-updating it with new data, and
4. when one is comfortable with the model, presenting the posterior and posterior predictive distributions in a way useful to answer the scientific questions at hand.

The mathematics of this process are elegant and easy. They only involve probability theory, using Bayes theorem, conditioning and marginalization.

Under the Bayesian framework, approach parameters θ , and observables $\tilde{y}$, are treated symmetrically through their joint probability distribution:

1. All questions about θ are answered through the posterior $\pi(\theta|y)$.
2. All questions about $\tilde{y}$ are answered through the posterior predictive $p(\tilde{y}|y)$.

Bayesians do not require complicated concepts like estimator bias, confidence level, p -value, or likelihood ranking to measure evidence in favor of an hypothesis or prediction. Instead, they can assign a probability to any hypotheses or future observable.

While Bayesian methods require a prior distribution to begin with, they are ultimately easier to explain, implement, and interpret than their non-Bayesian counterpart.

